

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
EEE 508 (OPTICAL COMMUNICATION) TEST 2nd SEMESTER (2023/2024) ANSWER
ALL QUESTIONS Time: 1 Hr

- 1 (a) With the aid of a labelled diagram, explain the transmission of information in an optical fiber communication system.
 (b) Give four drawbacks of fiber optic transmission

- 2 (a) Calculate the ratio of the stimulated emission rate to the spontaneous emission rate for an incandescent lamp operating at a temperature of 1000K. It may be assumed that the average operating wavelength is $0.5\mu\text{m}$.
 (b) Describe the following injection laser structures: (i) Gain-guided laser, (ii) Index-guided laser

- 3 (a) Differentiate between a p-n junction photodiode and a pin photodiode.
 (b) A photodiode has a quantum efficiency of 65% when photons of energy $1.5 \times 10^{-19}\text{J}$ are incident upon it.
 (i) At what wavelength is the photodiode operating?
 (ii) Calculate the incident optical power required to obtain a photocurrent of $2.5\mu\text{A}$ when the photodiode is operating as described above.



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SECOND SEMESTER EXAMINATION (2023/2024) Session

EEE 508: (Optical Communication)

Unit: 2

Instruction: ANSWER any three (3) questions in all

Time: 2 Hours

- 1 (a) A planar LED is fabricated from gallium arsenide which has a refractive index of 3.6.
- (i) Calculate the optical power emitted into air as a percentage of the internal optical power for the device when the transmission factor at the crystal-air interface is 0.68
- (ii) When the optical power generated internally is 50% of the electric power supplied, determine the external power efficiency (The refractive index for air is 1)
- [6 marks]
- (b) With the aid of a well labelled diagram, describe the physical structure of an optical fiber
- [9 marks]
- (c) Give five applications of optical fiber communication
- [5 marks]
- 2 (a) A photodiode has a quantum efficiency of 65% when photon of energy $1.5 \times 10^{-19} \text{ J}$ are incident upon it.
- (i) At what wavelength is the photodiode operating?
- (ii) Calculate the incident optical power required to obtain a photocurrent of $2.5 \mu\text{A}$ when the photodiode is operating as described above.
- ($h = 6.626 \times 10^{-34} \text{ Js}$, $e = 1.602 \times 10^{-19} \text{ C}$)
- [6 marks]
- (b) Explain the operation of an optical isolator and circulator using a diagram [6 marks]
- (c) Explain the bus and star topologies in an optical fiber LAN network and differentiate between them.
- [8 marks]
- 3 (a) A four-port multimode fiber FBT coupler has $60 \mu\text{W}$ optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004 , 26.0 and $27.5 \mu\text{W}$ respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device.
- [4 marks]
- (b) (i) What is Circuit and Packet switching?
- [4 marks]
- (ii) Compare Circuit and Packet switching
- [5 marks]
- (c) Describe the following types of amplifiers: (i) Semiconductor optical amplifier
- (ii) Erbium doped fiber amplifier
- [4 marks]
- (iii) Differentiate between an optical amplifier and a regenerative repeater
- [3 marks]

$$R = \frac{P_o}{h f}$$

$$A = 0.65$$

$$E = 1.602 \times 10^{-19} \text{ J}$$

$$h = 6.626 \times 10^{-34} \text{ Js}$$

$$1.5 \times 10^{-19}$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$0.68 \text{ A/m}^2$$

$$P = \frac{P_o}{P_i}$$

$$P_o = \frac{25 \times 10^{-5}}{0.68} = 3.60 \text{ mW}$$

$$\text{Split ratio} = \frac{P_3}{P_{\text{total}}} \times 1000 = 48.5 \%$$

is based on conventional fiber structure

$$E_{\text{loss}} = 10 \log \left(\frac{P_i}{P_o} \right)$$

$$= 10 \log$$

$$0.546$$

$$P_i = 60 \text{ mW}$$

$$P_o = 0.004 \text{ mW}$$

$$P_3 = 26.0 \text{ mW}$$

$$P_4 = 27.5 \text{ mW}$$

$$\frac{P_i}{P_o} = 3.34$$

face

4. (a) A transverse electro-optic modulator uses a KD *P Pockels cell of 5cm length having a width of 0.5cm. A light beam operating at a wavelength of 550nm is allowed to pass through the material. Calculate the half-wave voltage of the modulator. Assume the electro-optic coefficient of the KD *P crystal to be $26.5 \times 10^{-12} \text{ mV}^{-1}$ at the operating wavelength. Given that the refractive index KD *P cell is 1.51 [4 marks]
- (b) What is Optical Regeneration? How does a 3R Regenerator work? [6 marks]
- (c) What is FDM? State three advantages and disadvantages of Frequency Division Multiplexing [10 marks]
5. (a) (i) Explain the process of modulation and demodulation [4 marks]
- (ii) Why do we need to modulate? [4 marks]
- (b) Describe the following LED structures: (i) Planar LED (ii) Dome LED [8 marks]
- (c) What are the functions of an optical cable [4 marks]

Answer 95% for on modulator with
note on optical cable & electrical side

Good Luck



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B. ENG. DEGREE EXAMINATIONS. SECOND SEMESTER 2024/2025 SESSION

COURSE CODE: EEE508 COURSE TITLE: OPTICAL COMMUNICATION

UNITS: 2

TIME ALLOWED: 2 HOURS

Instruction: Answer Question One (1) and any other Two (2) Questions. All questions are Twenty (20) marks each.

QUESTION ONE

1. What effect does dispersion cause?
 - a. Scattering of light out the sides of the fiber
 - b. Stretching of signal pulses that increases with distance
 - c. Shrinking of signal pulses that become shorter with distance
 - d. Attenuation of signal pulses
2. Total internal reflection guides light along the core of an optical fiber, which has a higher refractive index than the surrounding cladding. True or False
3. Light can be polarized in _____ or _____ directions.
4. Graded-index fibers have standard core diameters of _____ or _____ micrometer.
5. The larger the fiber core, the more easily it can collect light from a light source. True or False
6. Fiber attenuation is the sum of _____ and absorption, measured together in decibels per kilometer.
7. Nonlinear effects are interactions between light waves, which generates noise and _____.
8. Temperature stabilization is important in maintaining stable wavelength and _____ power from the laser.
9. Fresnel reflection causes a _____ dB loss if there is an air gap between fiber ends.
10. Connectors hold fiber ends together in a _____ connection.
11. Splices have _____ attenuation than connectors.
12. Standard -form connectors include the snap-in SC and _____ ST and FC.
13. Wavelength -division multiplexing (WDM) Filters transmit selected _____ and reflect others.
14. Standard spacings for dense-WDM are 200, 100 and _____ GHz.



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15. Bubble switches direct signals by moving _____ back and forth in liquid guides to the junction points of planar waveguides.
16. Granularity is the _____ of signals in a network.
17. Optical Isolators are devices that transmit light in _____ direction.
18. The operation of optical Isolators usually depends on materials called _____ rotators.
19. An optical circulator sends light in one direction through a _____ of ports.
20. Active couplers are _____ with two or more outputs.

QUESTION TWO

- (a) Explain Six (6) advantages of optical fibers over copper wires. ✓
- (b) With the aid of a well labeled diagram, define Numerical Aperture (NA), and show that $NA = n_0 \sqrt{2\Delta}$. ✓
- (c) Explain the underlisted geometrical structure and index profile of Optical fibers, using mathematical expressions.
- i. Step-Index Profile
 - ii. Power-Law Index Profile
 - iii. Gaussian Index Profile ✓

QUESTION THREE

- (a) Explain with equations the following characteristics of Light Emitting Diode (LED)
- i. Internal Quantum Efficiency
 - ii. External Quantum Efficiency
 - iii. Power Efficiency ✓
 - iv. Luminous Efficiency
- (b) Provide Five (5) strict performance requirements of a detector. ✓
- (c) With the aid of a diagram, explain the process of fusion splicing. ✓

QUESTION FOUR

- (a) Explain these Three (3) characteristics of a detector:



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i. Quantum Efficiency

ii. Responsivity

iii. Rise time

(b) Explain the underlisted properties of optical fibers:

i. Chromatic Dispersion

ii. Brillouin Scattering

iii. Four-Wave Mixing

iv Stimulated Raman Scattering

v. Spectral Variation

(c) Identify the type of coupler drawn in Figure 1. Hence, explain its working principle.

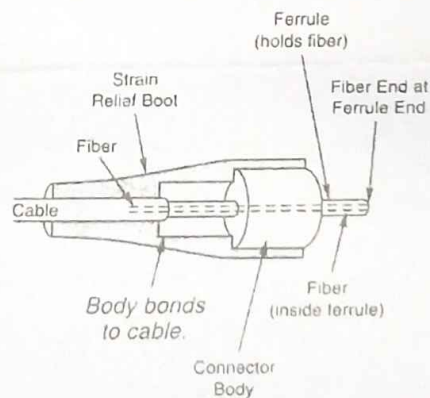


Figure 1

QUESTION FIVE

(a) With adequate diagrams, explain the working principles of:

i. Fiber Bragg grating

ii. Fused-Fiber couplers

iii. Equalizing Filter; as it relates to wavelength -division multiplexing

(b) Identify and provide a brief explanation of the switch shown in figure 2.



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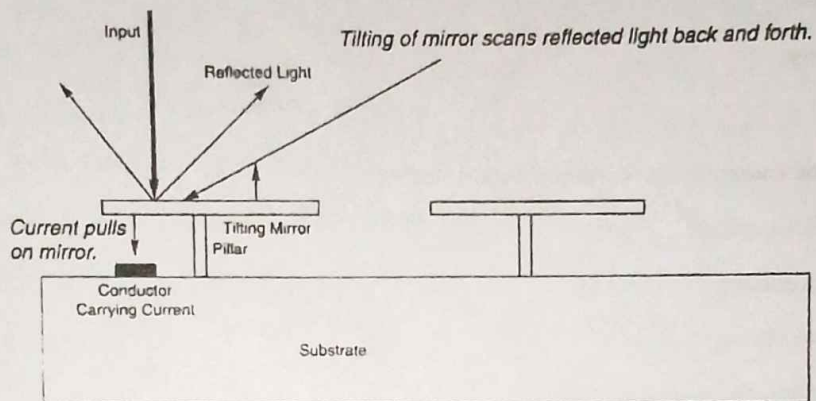


Figure 2

(c) Explain the following mechanism of Fiber-Optic Sensors;

- i. Polarization Sensing
- ii. Interferometric Sensing